

Technology Stack

Date	29/09/2026
Team ID	UNM1425COMICCRAFT
Project Name	COMICCRAFT
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML, CSS and Jinja2 templates	Simple, lightweight pages that work in any browser without extra tooling. Jinja2 is built into the FastAPI workflow, so the pages (index.html for input, comic_preview.html for the result, export_success.html for confirmation) are filled with the generated comic data on the server.
2	Backend / Server-Side	Python with FastAPI, served by Uvicorn (ASGI)	Python is the main language for AI libraries. FastAPI gives fast routing, Form(...) and Pydantic validation (PromptRequest), automatic API testing at /docs, and both form-based and JSON routes (/generate, /generate-comic/json). Uvicorn runs the app with auto-reload during development.

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3	Database / Data Storage	No database; file-based storage (static/panels and static/exports folders)	ComicCraft has no user accounts or saved records, so a database is not needed. Generated panel images are saved in static/panels and the exported PDFs in static/exports, which keeps the system simple and easy to run.
4	Cloud / Hosting / Deployment	Local deployment with Uvicorn in a Python virtual environment (venv)	Easy to set up and test with no hosting cost. The virtual environment keeps project packages separate from the system, and the app runs at http://127.0.0.1:8000 . It can later be moved to a cloud server or Docker if needed.
5	Version Control & CI/CD	Git	Lets the four team members track changes and work on modules (routes, AI functions, templates) together without overwriting each other's work. No automated CI/CD pipeline is used in the current version.
6	Third-Party APIs / Other Tools	• Google Gemini API (Gemini 1.5 Flash and Pro) • Hugging Face Diffusers with Stable Diffusion v1-5 • PyTorch, Transformers, Accelerate • FPDF • Pillow	Gemini Flash creates the structured 5-panel outline; Gemini Pro writes narration and dialogue. Stable Diffusion (via Diffusers, running on PyTorch) generates the comic-style illustration for each panel. FPDF builds the downloadable PDF, and Pillow handles image files.